Why our HTML Docs Don't Just `Print` and What to Do About It

Nikolaj Potashnikov

PhD in Economics, Solution architect, Course-IT

consulting@yandex.ru @nmpotashnikoff

> *The principal task of a conductor is not to put himself in evidence but to disappear behind his functions*
>
> *— Liszt Ferenc*

`Print`, `Export to Word`, `Export to PDF` are very often just a trap

What to do with a long line in a listing?

- We may scale
- Or use landscape orientation
- Or both, but would it be enough?
- If not, we may fire error for long lines or wrap them
- With linefeed and spaces? And how to copy?
- With indents? Still impossible to copy from PDF
- And on the web we can just add horizontal scroll bar

There is a core mismatch: semantic markup meets the rigid world of print

Iterations in converting simple text markup to print formats

- https://github.com/CourseOrchestra/course-doc: XSL-FO templates for Asciidoctor → DocBook backend
- https://github.com/CourseOrchestra/asciidoctor-open-document: Open Document Converter for Asciidoc
- https://github.com/fiddlededee/unidoc-publisher: UniDoc Publisher — any markup to any printing rendering engine

Main formats for printing

1. PDF
2. Text processing formats (Open XML — MS Office, Open Document — LibreOffice)
3. HTML?

CSS Paged Media — CSS extension, defining style specific for printing

Most widespread rendering approaches

- PDF ← native PDF-generating libraries
- PDF ← XSL-FO with FOP-processors
- PDF ← via TeX
- PDF ← HTML + Paged Media CSS
- DOCX/ODT, PDF ← +/- text processors (MS Word, LO Writer)

These technologies are not aligned in a great number of details like:

- Apache FOP has problems with Leader alignment (dots in a table of contents)
- LO Writer doesn't support typography (like keep with next) within table cells
- Microsoft doesn't recommend running automation tasks (like saving PDF) on a server

Some brief conclusions

Feel like speleologist?

- The world of printing is the world of constraints
- And those constraints differ for each technology, you often need to support several chains (exquisitely looking PDF with TeX and LibreOffice for coordination)
- With no universal solutions

UniDoc Publisher approach suits best if at least `one of`

- You don't prepare documentation especially for printing purposes
- You are automating documentation generation and hope it will look good, no matter what will be generated
- Your output format is one of the text processing format

In search for flexibility: Asciidoctor open document

Automation on the writer side

1. Asciidoctor parses markup into AST (Abstract Syntax Tree)
2. You may transform AST with Asciidoctor tree processor
3. Asciidoctor runs writer template for each AST node recursively
4. You may override writer with pure Ruby or with special Slim templates

A simplified processing AST example

```yaml
- !<OrderedList>
  roles:
  - "arabic"
  id: "ol-1"
  captioned_title:
      children:
      - !<Text>
        text: "Automation"
  children:
  - !<ListItem>
    children:
    - !<Paragraph>
      children:
      - !<Text>
        text: "Asciidoctor..."
  - !<ListItem>
...
```

```slim
- list_style = "#{get_basic_style} ordered-list"
- if captioned_title?
  text:p text:style-name="#{list_style}"
    text:bookmark text:name="#{id}"
    =captioned_title

text:list text:style-name="#{list_style}"
  - items.each_with_index do |item, index|
  ...
```

Great, but

- You can’t override part of a template
- You should invent something for styling

Styling? But text processors do support styling!

- `<span class="bold green">bold, green</span>` — impossible to apply two styles to one element

- Asciidoctor Open Document uses slightly extended intermediary OD format (to preserve AST attributes)
- It uses special functions that check, if style should be applied. It doesn’t know styling attributes but forces the Open Document style structure

And still

- Unexpectedly transforming this extended Open Document format became one of the most used features of Asciidoctor Open Document
- Styling as separate task of writing proved also to be useful
- Gradle was magnificent in gluing all parts together

Thoughts before the second step

- If creating universal converter is impossible…
- We should create **meta converter** – platform for building converters

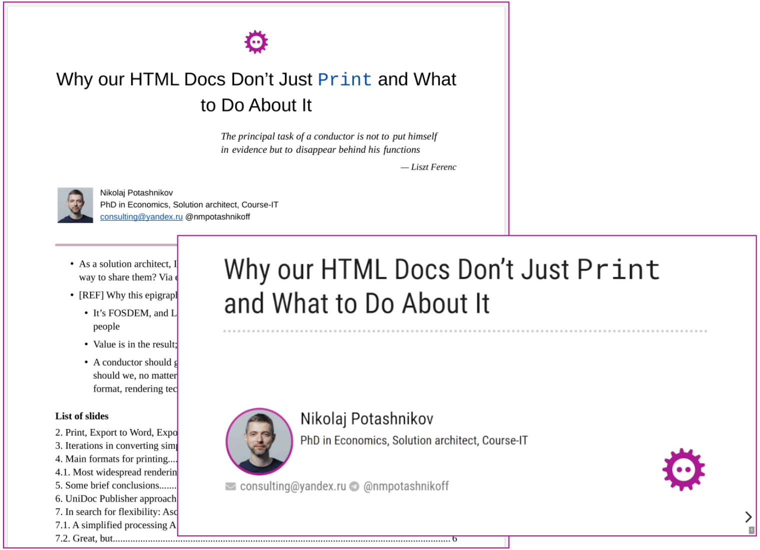
Estimated requirements

- Native converter as a reader
- Sound ways of transforming AST
- A good approach for styling as a separate focus
- Good integration with CI/CD with a focus on homogeneity

Native converter as a reader?

Each converter outputs HTML. HTML is quite semantic, why shouldn't we use it?

Let’s convert this presentation to LO Writer



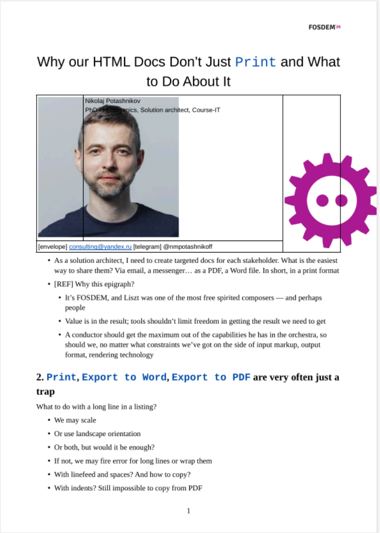
Notes on this demo

- All conversion settings are written in Kotlin
- Everything is in a single Gradle script (`build.gradle.kts`)
- The following code listings are excerpts from this script

Boilerplate

```
FodtConverter {
    html = AsciidocHtmlFactory()
        .getHtmlFromFile(File("${project.projectDir}/$presentationFile.adoc"), true)
    template = File("${project.projectDir}/template-1.fodt").readText()
    adaptWith(AsciidoctorOdAdapter)
    unknownTagProcessingRule = unknownTagProcessingRuleRevealJs()
    parse()
            // Processing AST and styling
    ast2fodt()
}
```

With no custom transformations and styling



FOSDEM'26

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[envelope] consulting@yandex.ru [telegram] @nmpotashnikoff

- As a solution architect, I need to create targeted docs for each stakeholder. What is the easiest way to share them? Via email, a messenger… as a PDF, a Word file. In short, in a print format
- [REF] Why this epigraph?
  - It's FOSDEM, and Liszt was one of the most free spirited composers — and perhaps people
  - Value is in the result; tools shouldn't limit freedom in getting the result we need to get
  - A conductor should get the maximum out of the capabilities he has in the orchestra, so should we, no matter what constraints we've got on the side of input markup, output format, rendering technology

2. `Print`, `Export to Word`, `Export to PDF` are very often just a trap

What to do with a long line in a listing?

- We may scale
- Or use landscape orientation
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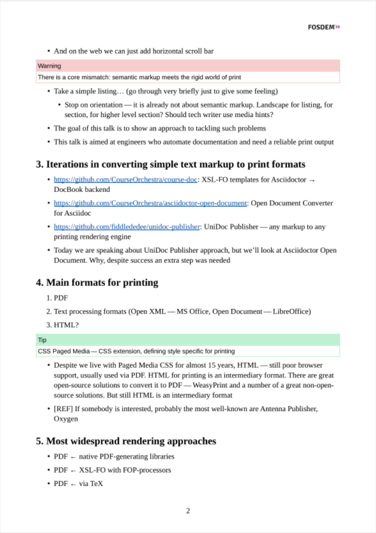
- With indents? Still impossible to copy from PDF

1



FOSDEM'26

- And on the web we can just add horizontal scroll bar

Warning
There is a core mismatch: semantic markup meets the rigid world of print

- Take a simple listing… (go through very briefly just to give some feeling)
  - Stop on orientation — it is already not about semantic markup. Landscape for listing, for section, for higher level section? Should tech writer use media hints?
- The goal of this talk is to show an approach to tackling such problems
- This talk is aimed at engineers who automate documentation and need a reliable print output

3. Iterations in converting simple text markup to print formats

- https://github.com/CourseOrchestra/course-doc: XSL-FO templates for Asciidoctor → DocBook backend
- https://github.com/CourseOrchestra/asciidoctor-open-document: Open Document Converter for Asciidoc
- https://github.com/fiddlededee/unidoc-publisher: UniDoc Publisher — any markup to any printing rendering engine
- Today we are speaking about UniDoc Publisher approach, but we'll look at Asciidoctor Open Document. Why, despite success an extra step was needed

4. Main formats for printing

1. PDF
2. Text processing formats (Open XML — MS Office, Open Document — LibreOffice)
3. HTML?

Tip
CSS Paged Media — CSS extension, defining style specific for printing

- Despite we live with Paged Media CSS for almost 15 years, HTML — still poor browser support, usually used via PDF. HTML for printing is an intermediary format. There are great open-source solutions to convert it to PDF — WeasyPrint and a number of a great non-open-source solutions. But still HTML is an intermediary format
- [REF] If somebody is interested, probably the most well-known are Antenna Publisher, Oxygen

5. Most widespread rendering approaches

- PDF ← native PDF-generating libraries
- PDF ← XSL-FO with FOP-processors
- PDF ← via TeX

2

Processing AST

```
ast().descendant { section ->
    section.sourceTagName == "section" &&
            section.descendant { it is Heading && it.level == 1 }
                .isNotEmpty()
}.first().also { it.insertBefore(makeTitle(it)) }.remove()
```

Rearranging title (Asciidoc source)

```
|===
a|
[.title-photo]
image::images/nmp1.jpg[]
a|
[.full-name]
Nikolaj Potashnikov

[.bio]
PhD in Economics, Solution architect, Course-IT
.2+>.>a|{nbsp}
[.logo]
image::images/fosdem-logo.svg[]
2+a|
[.contact]
icon:envelope[] consulting@yandex.ru icon:telegram[]{nbsp}@nmpotashnikoff
|===
```

Rearranging title, extracting semantics

```
val title = titleSlideSection.descendant { it is Heading && it.level == 1 }.first()
val notes = titleSlideSection.descendant { it.sourceTagName == "aside" }.first()
val (fullName, bio, photo, contact, logo) =
    arrayOf("full-name", "bio", "title-photo", "contact", "logo")
        .map { role -> titleSlideSection.descendant { it.roles.contains(role) }.first()
```

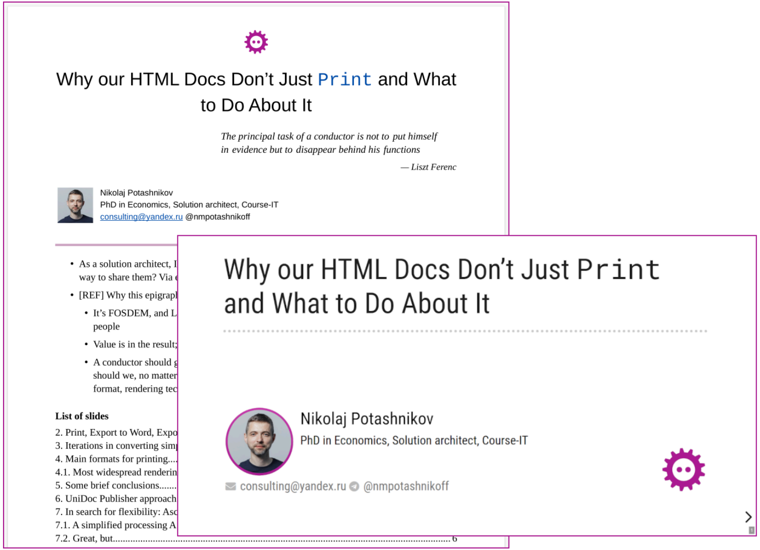
Rearranging/constructing title

```
appendChild(logo)
appendChild(title)
table {
    col(Length(18F)); col(Length(152F))
    roles("about-me")
    tableRowGroup(TRG.body) {
        tr {
            td { appendChild(photo) }
            td { arrayOf(fullName, bio, contact).forEach { appendChild(it) } }
        }
    }
}
appendChild(notes)
appendChild(Toc(2, "List of slides"))
normalizeImageDimensions()
```

Styling

```
OdtStyle { p ->
    if (p !is Paragraph) return@OdtStyle
    if (p.ancestor { it.roles.contains("logo") }.isEmpty()) return@OdtStyle
    attributes("style:master-page-name" to "First_20_Page")
},
OdtStyle { tableCell ->
    if (tableCell !is TableCell) return@OdtStyle
    if (tableCell.ancestor { it.roles.contains("about-me") }.isEmpty()) return@OdtStyle
    tableCellProperties {
        arrayOf("top", "right", "bottom", "left")
            .forEach { attributes("fo:border-$it" to "none") }
    }
},
```

Back to the result



Some more AST processing

```
ast().descendant { it.roles.contains("notes") }
    .forEach { it.insertBefore(HorizontalLine()) } 1
ast().descendant { it is Heading && it.level > 1 }
    .forEach {
        it.insertBefore(
            Paragraph().apply { roles("slide-finish") }
        )
    } 2
odtStyleList.add(odtStyles())
odtStyleList.add(rougeStyles()) 3
```

Extending AST

```kotlin
class HorizontalLine() : NoWriterNode() {
    override val isInline: Boolean get() = false
}
```

```kotlin
OdtCustomWriter { horizontalLine ->
    if (horizontalLine !is HorizontalLine) return@OdtCustomWriter
    preOdNode.apply {
        "text:p" {
            attributes("text:style-name" to "Horizontal Line")
            process(horizontalLine)
        }
    }
},
```

Testing

Content type	The result
paragraph	Paragraph 1 Paragraph 2

list	1. Item 1
	  1. Subitem
	    1. Subitem
	      1. Subitem
	2. Item 2

table	Subtable

Some paragraph after table.

Some paragraph after table.

Focus and trade-offs

- CI friendly – pure Gradle (or an ordinary Kotlin project) to rule them all
- No declarations, everything should be programmed
- Typed AST – check before run
- Clean and testable code
- One styling approach, but different styling API for each backend

Conclusion

- Treat printing as engineering: design it, test it, automate it
- Printing is a lossy transformation — some semantics cannot survive it
- Keep rendering logic programmable and under your control

Questions?

- https://github.com/fiddlededee/unidoc-publisher: UniDoc Publisher — any markup to any printing rendering engine
- https://github.com/fiddlededee/fosdem-printing: this presentation repository